Propositional Logic Encoding for Exam Scheduling

1 Problem Definition

We are tasked with scheduling a set of exams E into a set of time slots S.

- **Exams (E):** $\{M, P, C, CS, H\}$ corresponding to Mathematics, Physics, Chemistry, Computer Science, and History.
- **Slots (S):** $\{1, 2, 3\}$.

1.1 Variables

We define the propositional variables X_i for each exam $X \in E$ and slot $i \in S$:

$$X_i = \text{Exam } X \text{ is scheduled in slot } i$$

The total number of variables is $|E| \times |S| = 5 \times 3 = 15$.

2 Constraint Derivation

We seek a propositional formula Ψ such that a satisfying assignment to Ψ corresponds exactly to a valid schedule satisfying all constraints.

2.1 Fundamental Scheduling Constraints

Before encoding the specific rules of the problem, we must ensure the variables represent a well-formed function $f : E \to S$.

1. Completeness (At least one slot)

Every exam must be scheduled in at least one slot.

$$\phi_{\text{complete}} = \bigwedge_{X \in E} (X_1 \vee X_2 \vee X_3)$$

2. Uniqueness (At most one slot)

No exam can be scheduled in more than one slot.

$$\phi_{\text{unique}} = \bigwedge_{X \in E} \big(\neg(X_1 \wedge X_2) \wedge \neg(X_1 \wedge X_3) \wedge \neg(X_2 \wedge X_3)\big)$$

The base validity formula is:

$$\Phi_{\text{base}} = \phi_{\text{complete}} \wedge \phi_{\text{unique}}$$

2.2 Problem-Specific Constraints

Constraint 1: M and P Disjointness

"M and P cannot be in the same slot."
For every slot i, it is forbidden that both M_i and P_i are true.

$$\Phi_1 = \bigwedge_{i=1}^{3} \neg(M_i \wedge P_i)$$

Using logical equivalence $\neg(A \wedge B) \equiv (\neg A \vee \neg B)$, this can be written as:

$$\Phi_1 = (\neg M_1 \vee \neg P_1) \wedge (\neg M_2 \vee \neg P_2) \wedge (\neg M_3 \vee \neg P_3)$$

Constraint 2: C and CS Disjointness

"C must be in a different slot from CS."
Similarly, C and CS cannot share a slot.

$$\Phi_2 = \bigwedge_{i=1}^{3} \neg(C_i \wedge CS_i)$$

Constraint 3: Slot 1 Requirement

"At least one of M, P, C must be in S1."
This is a simple disjunction of assignments to slot 1.

$$\Phi_3 = M_1 \vee P_1 \vee C_1$$

Constraint 4: Conditional Constraint

"If H is in S1, then M cannot be in S3."
This is a material implication $H_1 \implies \neg M_3$.

$$\Phi_4 = H_1 \rightarrow \neg M_3$$

Converting to Clause Normal Form (CNF):

$$\Phi_4 \equiv \neg H_1 \vee \neg M_3$$

Constraint 5: Non-Consecutive Heavy Courses

"CS and M cannot be in consecutive slots."
Two slots i and j are consecutive if $|i - j| = 1$. The consecutive pairs are $(1, 2)$ and $(2, 3)$. We must forbid the configuration where one exam is in slot k and the other is in slot $k + 1$.

The forbidden configurations are:

- CS in 1, M in 2: $(CS_1 \wedge M_2)$
- M in 1, CS in 2: $(M_1 \wedge CS_2)$

- CS in 2, M in 3: $(CS_2 \wedge M_3)$
- M in 2, CS in 3: $(M_2 \wedge CS_3)$

The formula negates all these possibilities:

$$\Phi_5 = \neg(CS_1 \wedge M_2) \wedge \neg(M_1 \wedge CS_2) \wedge \neg(CS_2 \wedge M_3) \wedge \neg(M_2 \wedge CS_3)$$

3 Final Formula

The complete encoding Ψ is the conjunction of all derived formulas:

$$\Psi = \Phi_{\text{base}} \wedge \Phi_1 \wedge \Phi_2 \wedge \Phi_3 \wedge \Phi_4 \wedge \Phi_5$$

Writing this out fully:

$$\begin{aligned}
\Psi = &\left[\bigwedge_{X \in E} \left((X_1 \vee X_2 \vee X_3) \wedge \bigwedge_{1 \le j < k \le 3} \neg(X_j \wedge X_k) \right)\right] \quad \text{(Valid Schedule)} \\
&\wedge \left[\bigwedge_{i=1}^{3} \neg(M_i \wedge P_i)\right] \quad \text{(M, P Separate)} \\
&\wedge \left[\bigwedge_{i=1}^{3} \neg(C_i \wedge CS_i)\right] \quad \text{(C, CS Separate)} \\
&\wedge [M_1 \vee P_1 \vee C_1] \quad \text{(S1 Requirement)} \\
&\wedge [\neg H_1 \vee \neg M_3] \quad \text{(H and M Conditional)} \\
&\wedge [\neg(CS_1 \wedge M_2) \wedge \neg(M_1 \wedge CS_2)] \quad \text{(No Consecutive 1-2)} \\
&\wedge [\neg(CS_2 \wedge M_3) \wedge \neg(M_2 \wedge CS_3)] \quad \text{(No Consecutive 2-3)}
\end{aligned}$$

Every satisfying assignment of Ψ maps uniquely to a valid exam schedule.

Rigorous Proof via Natural Deduction

Problem Statement

We are asked to prove the following sequent:

$$((p \to \bot) \to \bot) \vdash p$$

using the specific set of inference rules provided:

1. $\to_i$: Implication Introduction
2. $\to_e$: Implication Elimination
3. $\bot_e$: Bottom Elimination (Ex Falso)
4. $(\to \bot)_e$: The specific rule $\frac{(\phi \to \bot) \to \psi \quad \phi \to \zeta \quad \psi \to \zeta}{\zeta}$

Derivation Strategy

We utilize the fourth rule, $(\to \bot)_e$. To make the rule result in our desired conclusion p, given our premise $((p \to \bot) \to \bot)$, we perform the following unification:

- Let $\phi = p$
- Let $\psi = \bot$
- Let $\zeta = p$

Substituting these into Rule 4, the required premises become:

1. $((p \to \bot) \to \bot)$ (This is our given assumption)
2. $p \to p$ (We must derive this)
3. $\bot \to p$ (We must derive this)

The Proof Tree

Below is the complete proof tree constructed using the `bussproofs` environment.

Step-by-Step Explanation

1. **Left Branch (Premise):** We begin with the given premise $((p \to \bot) \to \bot)$.
2. **Middle Branch (Identity):**
   - We assume p (marked with index 1).
   - We apply $\to_i$ to discharge the assumption, resulting in $p \to p$.
   - This satisfies the requirement $\phi \to \zeta$ (where $\phi = p, \zeta = p$).
3. **Right Branch (Ex Falso):**
   - We assume $\bot$ (marked with index 2).
   - By the rule $\bot_e$, from $\bot$ we can infer p.
   - We apply $\to_i$ to discharge the assumption, resulting in $\bot \to p$.
   - This satisfies the requirement $\psi \to \zeta$ (where $\psi = \bot, \zeta = p$).
4. **Conclusion:** With all three premises established, we apply rule $(\to \bot)_e$ to conclude ζ, which is p.

Propositional Logic Encoding of 4×4 Sudoku

1 Variables

Let $x_{i,j,k}$ be a Boolean variable where $i, j \in \{1, 2, 3, 4\}$ represent the row and column indices, and $k \in \{1, 2, 3, 4\}$ represents the value.

$$x_{i,j,k} = \begin{cases} \text{True} & \text{if cell } (i, j) \text{ contains value } k \\ \text{False} & \text{otherwise} \end{cases}$$

There are $4 \times 4 \times 4 = 64$ total variables.

2 The General Formula

The Sudoku puzzle is solvable if and only if the formula Φ is satisfiable.

$$\Phi = \Phi_{\text{cell}} \wedge \Phi_{\text{row}} \wedge \Phi_{\text{col}} \wedge \Phi_{\text{block}} \wedge \Phi_{\text{init}}$$

2.1 1. Cell Constraints (Φ_{cell})

Every cell (i, j) must contain exactly one number. This is the conjunction of "at least one" and "at most one".

$$\Phi_{\text{cell}} = \bigwedge_{i=1}^{4} \bigwedge_{j=1}^{4} \left[\left(\bigvee_{k=1}^{4} x_{i,j,k} \right) \wedge \left(\bigwedge_{1 \leq k < k' \leq 4} \neg (x_{i,j,k} \wedge x_{i,j,k'}) \right) \right]$$

2.2 2. Row Constraints (Φ_{row})

Every number k must appear at least once in every row i.

$$\Phi_{\text{row}} = \bigwedge_{i=1}^{4} \bigwedge_{k=1}^{4} \left(\bigvee_{j=1}^{4} x_{i,j,k} \right)$$

2.3 3. Column Constraints (Φ_{col})

Every number k must appear at least once in every column j.

$$\Phi_{\text{col}} = \bigwedge_{j=1}^{4} \bigwedge_{k=1}^{4} \left(\bigvee_{i=1}^{4} x_{i,j,k} \right)$$

2.4 4. Block Constraints (Φ_{block})

Every number k must appear at least once in every 2×2 subgrid. We define the four subgrids $B_{r,c}$ for $r, c \in \{0, 1\}$ where the cells are defined by row indices $\{2r+1, 2r+2\}$ and column indices $\{2c+1, 2c+2\}$.

$$\Phi_{\text{block}} = \bigwedge_{r=0}^{1} \bigwedge_{c=0}^{1} \bigwedge_{k=1}^{4} \left(\bigvee_{m=1}^{2} \bigvee_{n=1}^{2} x_{2r+m,2c+n,k} \right)$$

3 The Instance Formula (Φ_{init})

This component enforces the clues provided in the specific problem image. It is a conjunction of Unit Clauses (literals that must be true).

Based on the image provided:

$$\begin{aligned}
&\text{Row 1:} \quad x_{1,1,1} \wedge x_{1,3,3} \\
&\text{Row 2:} \quad x_{2,4,2} \\
&\text{Row 3:} \quad x_{3,4,3} \\
&\text{Row 4:} \quad x_{4,1,2} \wedge x_{4,4,1}
\end{aligned}$$

Thus, the specific formula for this instance is:

$$\Phi_{\text{init}} = x_{1,1,1} \wedge x_{1,3,3} \wedge x_{2,4,2} \wedge x_{3,4,3} \wedge x_{4,1,2} \wedge x_{4,4,1}$$

4 Summary

The complete propositional formula for this problem is the conjunction of all constraints defined above. A satisfying truth assignment to the variables $x_{i,j,k}$ corresponds to the unique valid solution of the puzzle.

CNF Encoding of the 4×4 Sudoku Instance

1 Variable Definition

Let $x_{i,j,k}$ be a Boolean variable representing that the cell at row i, column j contains value k.

- $i, j, k \in \{1, 2, 3, 4\}$
- Total variables: $4^3 = 64$.

2 The Clause Sets

The CNF formula is a conjunction of the following sets of clauses.

2.1 1. Cell Definedness (C_{def})

Every cell must contain **at least one** value.

$$\bigwedge_{i=1}^{4} \bigwedge_{j=1}^{4} (x_{i,j,1} \vee x_{i,j,2} \vee x_{i,j,3} \vee x_{i,j,4})$$

(16 clauses)

2.2 2. Cell Uniqueness (C_{unique})

Every cell must contain **at most one** value. For every cell (i, j) and every distinct pair of values u, v, if the cell contains u, it cannot contain v.

$$\bigwedge_{i=1}^{4} \bigwedge_{j=1}^{4} \bigwedge_{1 \leq u < v \leq 4} (\neg x_{i,j,u} \vee \neg x_{i,j,v})$$

(16 cells $\times$ 6 pairs = 96 clauses)

2.3 3. Row Constraints (C_{row})

Every row i must contain every value k (at least once).

$$\bigwedge_{i=1}^{4} \bigwedge_{k=1}^{4} (x_{i,1,k} \vee x_{i,2,k} \vee x_{i,3,k} \vee x_{i,4,k})$$

(16 clauses)

2.4 4. Column Constraints (C_{col})

Every column j must contain every value k (at least once).

$$\bigwedge_{j=1}^{4} \bigwedge_{k=1}^{4} (x_{1,j,k} \vee x_{2,j,k} \vee x_{3,j,k} \vee x_{4,j,k})$$

(16 clauses)

2.5 5. Subgrid (Block) Constraints (C_{block})

Every 2×2 block must contain every value k. The blocks $B_{r,c}$ (where $r, c \in \{0, 1\}$) cover cells $(2r + 1 \ldots 2r + 2, 2c + 1 \ldots 2c + 2)$.

$$\bigwedge_{r=0}^{1} \bigwedge_{c=0}^{1} \bigwedge_{k=1}^{4} \left(\bigvee_{m=1}^{2} \bigvee_{n=1}^{2} x_{2r+m,2c+n,k} \right)$$

(16 clauses)

3 Instance Specific Unit Clauses (C_{givens})

These clauses enforce the initial configuration given in the problem image. They are single-literal clauses (Unit Clauses).

$$\begin{aligned} C_{\text{givens}} = \quad & (x_{1,1,1}) \wedge (x_{1,3,3}) \\ & \wedge (x_{2,4,2}) \\ & \wedge (x_{3,4,3}) \\ & \wedge (x_{4,1,2}) \wedge (x_{4,4,1}) \end{aligned}$$

(6 clauses)

4 Total Complexity

The complete CNF encoding for this specific instance consists of:

$$16 + 96 + 16 + 16 + 16 + 6 = 166 \text{ Clauses}$$

Any Boolean Satisfiability (SAT) solver that finds a variable assignment satisfying all 166 clauses has found the unique solution to the puzzle.

Rigorous Proof of Logical Validity

1 Problem Statement

Let p_1, p_2, p_3, q be propositional variables. We wish to prove that the following logical implication is a tautology:

$$[(p_1 \to q) \land (p_2 \to q) \land (p_3 \to q)] \longrightarrow [(p_1 \lor p_2 \lor p_3) \to q] \tag{1}$$

2 Definitions and Axioms

The proof relies on the following standard logical equivalences (Laws of Boolean Algebra). Let A, B, C be arbitrary propositions.

1. **Material Implication:** $A \to B \equiv \neg A \lor B$
2. **Distributive Law:** $(A \lor C) \land (B \lor C) \equiv (A \land B) \lor C$
   (Generalized: $\bigwedge_i (A_i \lor C) \equiv (\bigwedge_i A_i) \lor C$*)*
3. **De Morgan's Law:** $\neg A \land \neg B \equiv \neg(A \lor B)$
   (Generalized: $\bigwedge_i \neg A_i \equiv \neg(\bigvee_i A_i)$*)*
4. **Reflexivity of Implication:** $A \to A$ is a tautology ($\top$).

3 Formal Proof

Theorem 1. *The proposition Φ defined in Eq. (1) is a tautology.*

Proof. Let logical equivalence be denoted by $\equiv$. We define the antecedent (Left Hand Side) of the main implication as L. We will transform L algebraically to show it is equivalent to the consequent (Right Hand Side).

Let L be:

$$L = (p_1 \to q) \land (p_2 \to q) \land (p_3 \to q)$$

Step 1: Eliminate Implications

Using the rule of *Material Implication* ($A \to B \equiv \neg A \lor B$), we rewrite each term:

$$L \equiv (\neg p_1 \lor q) \land (\neg p_2 \lor q) \land (\neg p_3 \lor q)$$

Step 2: Apply Distributivity

We observe that the variable q is common to all disjunctions. By the *Distributive Law* (specifically, distribution of $\lor$ over $\land$), we can factor out q:

$$L \equiv (\neg p_1 \land \neg p_2 \land \neg p_3) \lor q$$

Step 3: Apply De Morgan's Law

We apply *De Morgan's Law* to the conjunction of negations $(\neg p_1 \wedge \neg p_2 \wedge \neg p_3)$. This allows us to pull the negation outside:

$$(\neg p_1 \wedge \neg p_2 \wedge \neg p_3) \equiv \neg(p_1 \vee p_2 \vee p_3)$$

Substituting this back into our expression for L:

$$L \equiv \neg(p_1 \vee p_2 \vee p_3) \vee q$$

Step 4: Reintroduce Implication

Let $P = (p_1 \vee p_2 \vee p_3)$. Our expression is now $\neg P \vee q$. Using the rule of *Material Implication* in reverse $(\neg A \vee B \equiv A \to B)$:

$$\neg(p_1 \vee p_2 \vee p_3) \vee q \equiv (p_1 \vee p_2 \vee p_3) \to q$$

Conclusion

We have demonstrated through a chain of equivalences that:

$$\underbrace{((p_1 \to q) \wedge (p_2 \to q) \wedge (p_3 \to q))}_{\text{Antecedent}} \equiv \underbrace{((p_1 \vee p_2 \vee p_3) \to q)}_{\text{Consequent}}$$

The original formula is of the form $L \to L$. By the law of identity (Reflexivity), $L \to L$ is always True.

Therefore, the formula is a tautology. $\square$